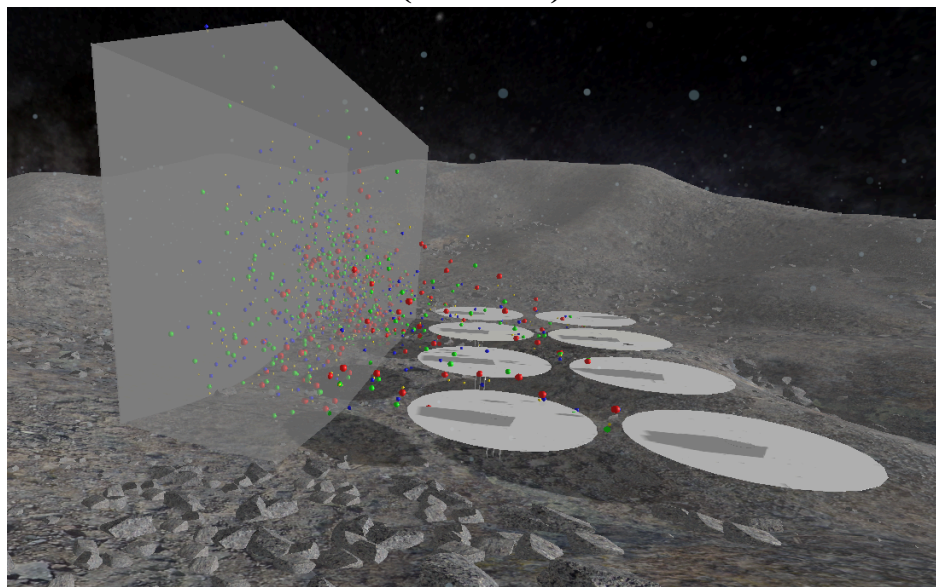




2025 RASC-AL Technical Paper

Theme: Lunar Surface Power and PMAD Architectures Theme
Power Energy Transfer Architecture for the Lunar surface
(PETAL)



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Project PETAL

University of Hawai'i at Manoa with University of Hawai'i at Hilo

Theme: Lunar Surface Power and Power Management and Distribution (PMAD) Architectures

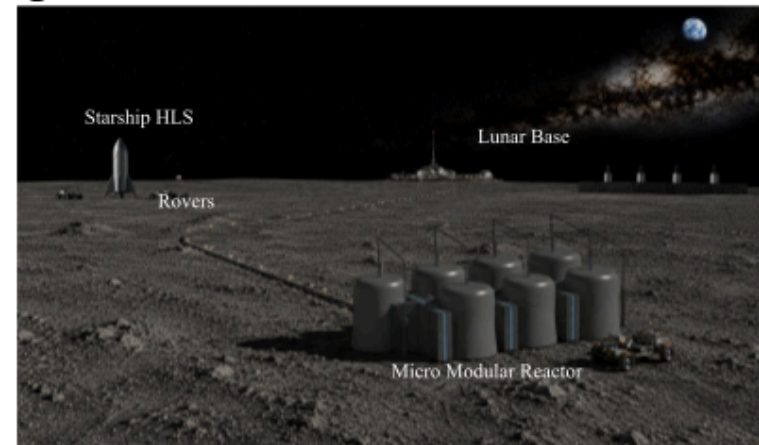


RASC-AL
Revolutionary Aerospace Systems Concept Academic Linkage

Objectives & Technical Approach:

- Power a lunar South Pole base through all three Artemis phases, from early robotic ops to a permanent 120 kW base
- Survive the 336-hour lunar night with a buried regolith thermal vault and Brayton Cycle turbines - no imported batteries
- Eight small fission reactors share one shielding wall instead of carrying individual shields, cutting launched mass by 30–40%
- Power travels 10 km over a buried HVDC line to five standard voltage tiers, all using the same universal docking port
- Built entirely by robots before any crew arrives, with reactors hot-swappable without ever shutting the grid down

Image:



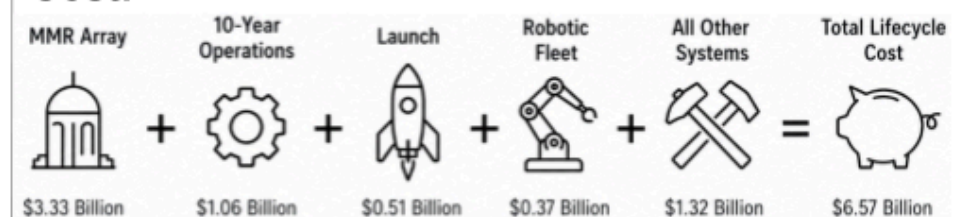
Key Design Details & Innovations:

- Launched mass is ~38,500 kg with ~639,800 kg built from moon dirt on-site; runs 120 kW continuously, peaks at 160 kW during the day, all safety decisions made autonomously
- The shared reactor wall grows with the base, every new reactor slots in behind the same wall with no extra shielding needed
- Every load on the base uses the same docking port, so future missions plug straight in without redesigning anything
- The vault - 604,800 kg of thermal mass - is built entirely from local regolith, cutting launched mass dramatically and proving out the same construction approach planned for Mars
- Almost every subsystem transfers to Mars with only minor modifications, using the Moon as a proving ground so a flight-tested system arrives at Mars instead of an unvalidated one

Schedule:



Cost:



2026 RASC-AL Technical Paper Compliance Matrix

Lunar Surface Power and PMAD Architectures

General Technical Paper Information	Y/N
Does your submission adhere to the requirements and constraints of the selected topic and the design competition? Does it adhere to the page limitations and formatting guidelines?	Y
Have you proposed synergistic applications of NASA's planned current investments with new innovative approaches / capabilities / technologies to support your proposed solution?	Y
Does your technical paper present original engineering analysis of innovative approaches, capabilities and/or new technologies for evolutionary architecture development to enable future missions, reduce cost, and/or improve safety?	Y
Have you provided sufficient justification for your assumptions?	Y
Have you addressed reliability and human safety considerations when trading various design options?	Y
Is your approach guided by sound systems engineering and architectural trades?	Y
Have you documented and justified your technology readiness levels (TRLs) for the technologies associated with your proposed solution?	Y
Are appropriate assessments of costs for technology maturation, system development, and production & operations included in your technical paper?	Y
Did you include an appropriate project schedule and test plan with realistic development and annual operating costs (i.e., budget for technology maturation, system development, and production & operations)?	Y
Lunar Surface Power and PMAD Architectures Theme	Page #
Where does your technical paper describe the relevant primary power generation systems, energy storage systems, and power management and distribution functions?	4-6
Where does your technical paper describe the different needs and use cases your architecture can support?	1, 3, 7, 12
Where does your technical paper describe the infrastructure for providing connectivity between power generation, energy storage, and client systems?	3, 7, 17
Where does your technical paper describe the interface requirements your architecture imposes on other systems?	11
Where does your technical paper describe how your architecture is extensible to supporting future human operations on Mars?	16

1. Problem Statement and Background of PETAL

Sustaining a human presence at the lunar South Pole demands a power infrastructure that evolves across three distinct phases: robotic site preparation requiring 1-10 kW; initial crewed operations requiring 10-40 kW; and a permanent base requiring 40-120+ kW continuous. The challenge is not simply generating power - it is building an integrated, evolvable system that delivers power reliably across the 336-hour lunar night, survives electrostatically charged regolith and extreme thermal cycling, and supports entirely different client loads without redesigning the power plant at each step.

The South Pole is the Artemis target because ridge crests near Shackleton and de Gerlache craters receive greater than 80% solar illumination and permanently shadowed regions hold accessible water ice [25, 26]. But those ridges also present the harshest operational environment in the inner solar system: surface temperatures swinging from 100 K to 400 K, sub-micron charged regolith particles that defeat conventional connectors, ionizing radiation from both the lunar environment and on-site reactor operations, and a 336-hour night that no solar array can bridge alone [2, 17, 21].

Current state-of-the-art lunar power concepts treat generation, storage, and distribution as separable problems, producing point solutions that each collapse under the combined demands of the full operational lifecycle. Project PETAL (Power and Energy Transfer Architecture for the Lunar Surface) takes the integrated system view. Every interface is standardized. Every critical path is redundant. Every subsystem maps onto a NASA development program already in progress - FSP fission [18, 23], TYMPO HVDC [31], RASSOR/IPEx excavation [37], and SEER sintering [36] - and every technology choice was evaluated simultaneously on lunar merit and Mars extensibility.

The architecture must deliver five things simultaneously: scalable power from 100 W to 120+ kW; 336-hour night survival without resupply; modular growth across multiple Artemis missions; fault tolerance that survives any single component failure; and an interface standard so simple that future mission planners never need to redesign the bus. PETAL delivers all five.

2. Innovations and Changes of the Proposal

2.1 Innovative Approaches and Key Capabilities

PETAL advances the state of the art across five specific innovations, each of which addresses a known limitation of existing lunar power concepts.

Shared Shielding Wall. Rather than shielding each of the eight FSP Micro Modular Reactors (MMRs) independently - which would require 6,400–9,600 kg of individual shadow shielding - PETAL deploys a single 5 m tall $\times$ 12 m long $\times$ 3 m thick wall of regolith–polyethylene composite constructed in-situ by the SEER end-effector [20, 22, 36]. The shared-wall architecture saves between 34% (four-reactor baseline) and 61% (eight-reactor design point) of shielding mass relative to individually shielded modules (Appendix A.4); the design-of-record eight-reactor configuration realizes the upper bound of this range, subject to a layout-driven uncertainty of approximately $\pm 5\%$ versus independent shields, while creating standing infrastructure that absorbs every future reactor delivered to the site with zero incremental shielding cost. Per-unit launch mass without individual shield: approximately 400 kg. Shared wall launched hardware: approximately 2,300 kg. The remaining bulk ($\sim 288,000$ kg) is sintered in-situ from local regolith.

Sintered-Regolith Thermal Vault with Dual-CBC Recovery. A lithium-ion battery bank sized for 13,440 kWh of night energy at 40 kW would mass approximately 58,000 kg and operate outside the thermal range of every space-qualified Li-ion chemistry. PETAL instead uses a sintered-regolith thermal vault: a 7.75 m diameter $\times$ 8 m deep cylindrical excavation whose walls and floor are sintered to $k \approx 1.5$ W/m·K (from $k \approx 0.001$ W/m·K loose dust [3]), charged by 240 kWt of MMR waste heat via sodium heat pipes, and discharged through dual Closed Brayton Cycle turbines at approximately 30% efficiency. Vault-only steady-state margin against the 13,440 kWh design night requirement is $1.50\times$ (67,200 kWh thermal $\times$ 30% = 20,160 kWh electrical). First-cycle margin - when only the diffusion shell heats during initial charging - is $1.29\times$. With full vault charging and continuous MMR contribution of the night load, effective survival margin reaches $\sim 2.7\times$ under nominal conditions. Vault structural mass is entirely in-situ regolith - zero launched storage mass for 67,200 kWh of thermal capacity [4, 34].

Heritage-Anchored 10 kV HVDC Backbone. PETAL transmits power at 10 kV DC over 10 km, achieving 94.7% efficiency from reactor to the 3 kV habitat-side distribution bus through two GaN HEMT conversion stages and 50 mm² copper conductor (full calculation in Appendix A.3). End-to-end

efficiency from reactor to load - including final tier-step-down losses (97% per stage) - is 91.9% to AC tiers and 90.5% to low-voltage DC tiers. HVDC was chosen specifically because seven decades of terrestrial commercial operating data make every cost, loss, and failure-mode estimate defensible. The NASA TYMPO program is qualifying this technology for lunar burial [31]. Buried conduit provides both dust and thermal-cycling protection with no novel technology required.

IDSS-Derived Modular Docking Interface. Every load on the PETAL grid - from a 100 W science instrument to a 60 kW ISRU plant - connects through the same standardized docking port, derived from the International Docking System Standard [44]. The port enforces voltage-class matching (five tiers: 24 VDC, 28 VDC, 48 VDC, 120/240 VAC, 480 VAC), $\pm 10\%$ tolerance, dust-tight dual-stage sealing (outer elastomeric wiper + inner hermetic O-ring, replacing the inapplicable IP67 water-ingress rating with Apollo and MISSE-qualified lunar connector practice), CAN-class load-reporting handshake, and beveled LiDAR-fiducial lead-in for ± 2 mm rover-arm insertion (GITAI S2 ISS-demonstrated heritage [41]). New Artemis missions connect without bus redesign.

Robotic-First Lifecycle with Sodium Freeze-Seal Hot-Swap. All excavation, sintering, heat pipe installation, and HVDC routing is completed by autonomous and teleoperated rovers months before any crew arrives. The sodium heat pipe freeze-seal mechanism — a passive cold-trap annular fin radiator at a 2.0 m standoff from the reactor face, drawing on 30 years of EBR-II sodium reactor operations [Koch 2008] and the SP-100 space nuclear program [Mondt 1994] — allows individual MMR replacement without grid shutdown. Each reactor-side heat pipe uses an internal titanium wick structure to maintain passive capillary return of the liquid sodium working fluid across the 800–900 K operating range. After five days, decay heat drops to 193 W (ANS-5.1-2014), the cold-trap fin passively freezes the sodium plug within 6–12 hours, and the rover disconnects and replaces the reactor while the remaining seven units carry the load. Freeze-seal sensitivity analysis under $\pm 20\%$ decay heat and $\pm 10\%$ emissivity uncertainties is provided in Appendix A.9.

2.2 Changes Since Proposal

Six aspects of PETAL have matured since the proposal submission:

- (1) Heat pipe fluid: MMR working fluid corrected from titanium–water to sodium for 800–900 K operating temperatures, leveraging EBR-II and SP-100 heritage (§3.3).
- (2) Vault geometry: Reduced to 7.75 m diameter $\times$ 8 m depth based on derated RASSOR throughput, correcting first-cycle night survival margin to $1.29\times$ and aligning excavation timeline with Stage 1 power constraints (§3.4, §3.6).
- (3) Dual-CBC N+1: Reliability enhanced with two 20 kWe CBC units in parallel, eliminating single-string failures for the 40 kW essential night load (§3.4).
- (4) Freeze-seal update: Mechanism revised from active thermoelectric to passive cold-trap fin radiator, ensuring sodium solidification against 193 W of decay heat without active hardware or grid power (§2.1, §3.6).
- (5) HVDC and PMAD: Transmission efficiency reconciled to 94.7% reactor-to-3 kV-bus; PMAD software classified NASA Class B with hardware interlock relay for life-support bus per ISS EPS philosophy [JSC48793] (§3.5, Appendix A.3).
- (6) Risk Register and terminology: Three risks added (R9 shared-wall localized damage, R10 PMAD hub physical destruction, R11 FLEX-Cabling rover loss); terminology standardized to "Development Phase" for NASA lifecycle stages and "Deployment Stage" for on-Moon sequences (§3.7).

3. Project Overview and Mission Design

3.1 Site Selection

PETAL is sited on the Connecting Ridge between Shackleton Crater and de Gerlache Crater on the lunar South Pole, a region identified as one of the few locations that simultaneously satisfies power, thermal, resource, and safety constraints. First, illumination duration: ridge crests in this corridor receive solar illumination for greater than 80% of the lunar synodic month, with multi-day continuous illumination periods on the order of 200+ hours during polar summer [25, 26]. Second, thermal stability: surface temperatures along the ridgeline cycle through a markedly

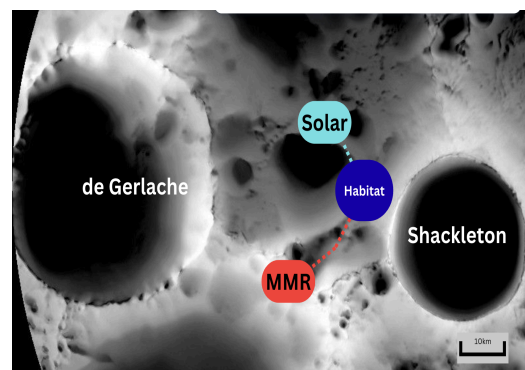


Figure 1. Top-view map

narrower range than equatorial sites, reducing fatigue stress on cabling, structural mounts, and HVDC insulation [25]. Third, proximity to permanently shadowed regions within Shackleton and adjacent craters place water ice and therefore future ISRU within tens of kilometers of the power plant. Fourth, the corridor's geometry permits the protected nuclear zone (the eight-MMR array, the shielding wall, and the thermal vault) to be sited at least 1 km from the planned crew habitat zone, satisfying the radiation safety standoff distance derived from FSP studies [27].

3.2 System Architecture Overview

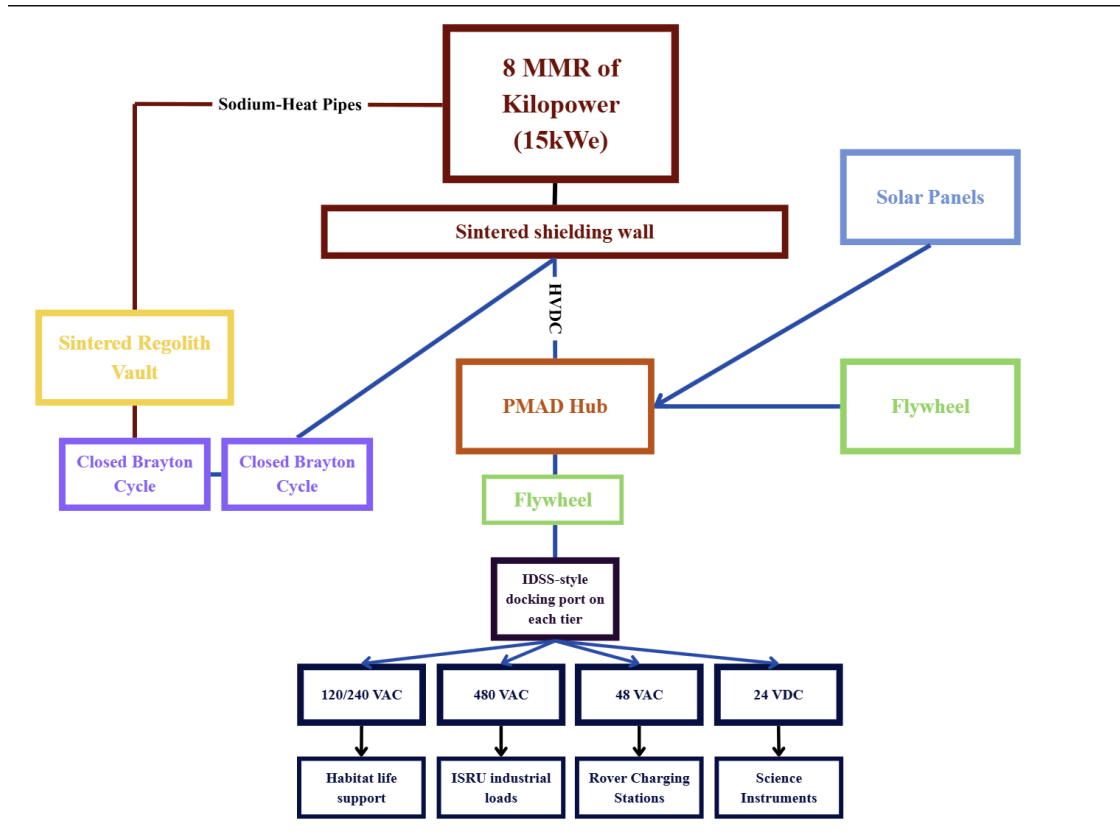


Figure 2. PETAL System Architecture Overview - power flow from MMR array through HVDC backbone to PMAD hub and distribution tiers.

PETAL combines eight 15 kWe MMRs, 40 kW supplemental vertical solar, a sintered-regolith thermal vault, dual flywheels, a 10 kV HVDC backbone, and a habitat-side PMAD hub. Power is generated in the protected nuclear zone, transmitted 10 km to the habitat zone, stepped down into standardized voltage tiers, and distributed through IDSS-derived docking interfaces. Detailed generation, storage, transmission, and deployment logic are described in §§3.3-3.6.

3.3 Power Generation

PETAL's primary generation system is an eight-unit MMR array providing 120 kWe of continuous baseload power, with each unit producing approximately 15 kWe and rejecting 30 kWt of waste heat for thermal-vault charging.

The selected architecture is justified through a reactor trade study comparing a single 120 kWe reactor, eight independently shielded reactors, an RTG array, and the PETAL approach of eight reactors sharing one shielding wall. The PETAL configuration provides the

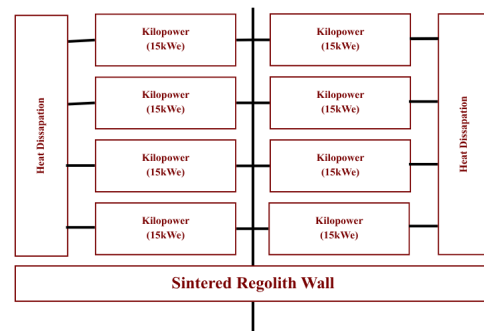


Figure 3. Power flow from MMR array through HVDC backbone

strongest balance of redundancy, maintainability, scalability, and shielding mass efficiency. As detailed in §2.1, the shared regolith-polyethylene shielding wall reduces repeated per-reactor shielding mass while creating permanent infrastructure for future reactor additions.

Within the protected nuclear zone, the eight MMRs are arranged in a compact two-row by four-column layout approximately 1 km from the crew habitat. Crew-facing protection is provided by the shared wall and regolith overburden described in §2.1, while secondary internal partitions reduce cross-radiation between reactor units and protect nearby sensors, cabling, control electronics, power conversion hardware, and maintenance interfaces. compartment separated by 60 cm composite partitions consisting of 15 cm sintered regolith, 30 cm borated polyethylene, and 15 cm sintered regolith. Final partition geometry will be refined through coupled radiation and thermal modeling to preserve heat rejection, heat-pipe routing, and rover access.

Thermal energy from the MMR array is transferred through a high-temperature sodium heat pipe network. Sodium is selected as the working fluid because the reactor-to-vault thermal transport loop operates near 800–900 K, where water heat pipes are no longer appropriate. This network routes rejected reactor heat into the sintered-regolith thermal vault for later recovery during lunar night operations.

Supplemental daytime power is provided by vertically mounted solar arrays positioned along the Connecting Ridge. During illuminated periods, solar generation offsets a portion of habitat and industrial electrical demand, allowing more reactor thermal output to charge the thermal vault while reducing transient loading stress on the MMR array.

Short-duration power fluctuations are managed by the dual-flywheel energy storage system integrated into the PMAD bus. The flywheels absorb transient load spikes and generation dips, stabilizing bus voltage and reducing dynamic stress on the MMR and solar subsystems. Detailed flywheel sizing and lunar-night storage roles are discussed in §3.4.

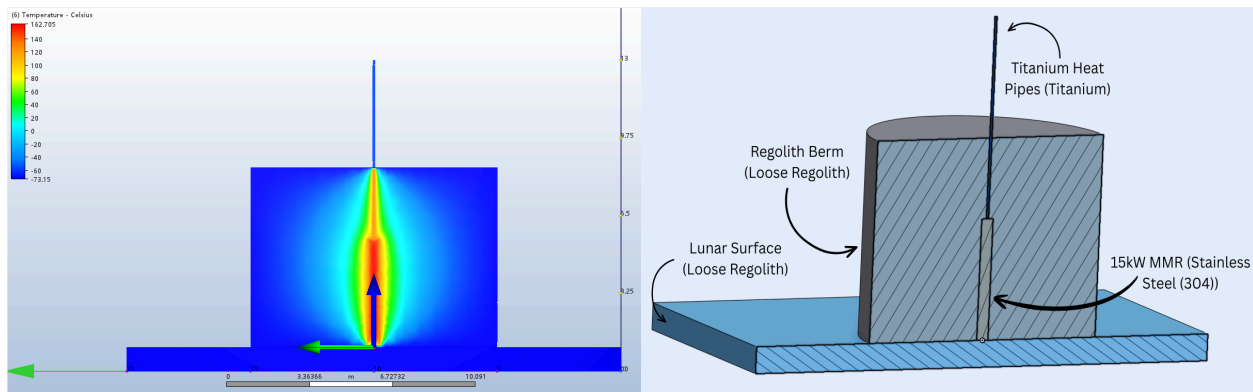


Figure 4. MMR Regolith Burial Thermal Simulation. Figure 5. Outline of burial

Regolith Burial and Environmental Protection. Each MMR is buried beneath a 3 m loose-regolith berm constructed from excavated vault material during Stage 1. The burial provides passive protection against surface thermal cycling, micrometeoroid impact, radiation exposure, and mutual line-of-sight coupling between adjacent reactor units. A steady-state Autodesk CFD 2027 simulation modeled a single 15 kWe MMR as a stainless-steel cylinder rejecting 30,000 W of waste heat into a loose-regolith berm, with a titanium heat pipe routed to the surface and boundary conditions of 200 K ambient and 400 K at the heat-pipe exit. Results show that 3 m of regolith reduces the surface temperature above the reactor to ambient 200 K, while the reactor casing remains at approximately 435°C. Heat is fully contained within the berm, validating the burial design as a passive thermal, environmental, and infrastructure-protection measure while routing rejected heat to the TES vault.

3.4 Energy Storage and Lunar Night Operations

PETAL uses a layered energy storage architecture to survive extended lunar darkness. The long-duration storage element is a sintered-regolith thermal vault coupled to Closed Brayton Cycle (CBC) recovery, while the dual-flywheel system provides rapid stabilization during short-duration load spikes, generation dips, and fault ride-through events. This separates energy storage by time scale: the thermal

vault provides multi-day survival capability, while the flywheels stabilize second-to-minute-scale electrical disturbances. A storage trade study evaluated lithium-ion battery

banks, regenerative fuel cells, flywheel-only storage, and a sintered-regolith thermal vault paired with CBC recovery. Lithium-ion batteries were rejected because the mass required to support the 13,440 kWh lunar-night energy demand would be approximately 58,000 kg and would operate outside the qualified thermal range of every space-rated Li-ion chemistry. Regenerative fuel cells reduced battery mass but added reactant storage, plumbing complexity, and lower round-trip efficiency. Flywheel-only storage was eliminated because flywheels are optimized for high-power transient response rather than multi-day energy storage. PETAL therefore adopts the sintered-regolith thermal vault because it leverages in-situ material, scales with settlement demand, and avoids prohibitively large imported storage systems.

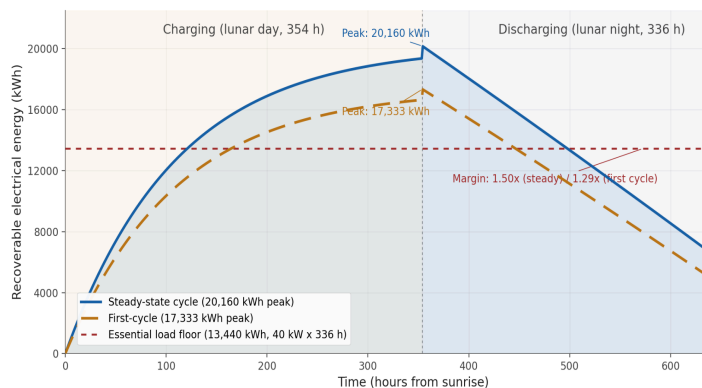


Figure 6. PETAL Thermal Vault: Stored Energy vs. Time

Thermal Vault Architecture. The thermal vault is a cylindrical excavation 7.75 m in diameter and 8 m deep, constructed using compacted sintered lunar regolith as the primary thermal mass. The excavation removes approximately 378 m³ of regolith, corresponding to 604,800 kg at an assumed bulk density of 1,600 kg/m³. Waste heat from the MMR array is transferred into the vault through high-temperature sodium heat pipes, raising the vault temperature from approximately 200 K to 800–900 K during charging operations. Detailed thermal-energy calculations are provided in Appendix A.1.

Vault Energy and Dark-Window Margin. At full charge, the vault stores approximately 67,200 kWh of thermal energy. At 30% nominal CBC recovery efficiency, this corresponds to 20,160 kWh of recoverable electrical energy. Against the 336-hour design night requirement of 13,440 kWh electrical (40 kW × 336 h), the vault provides a 1.50× steady-state vault-only margin. On the first lunar night after commissioning, when only the diffusion shell has fully heated, recoverable electrical energy is 17,333 kWh, yielding a 1.29× first-cycle margin. Under the more representative 168-hour ridge-crest dark window, the first-cycle margin increases to 2.58× and the steady-state margin approaches 3.0×. Under nominal operation, where the MMR array continues supplying essential loads while the vault serves as deep backup, effective survival margin exceeds 2.7×.

Vault Construction Sequence. The thermal vault requires excavation of approximately 604,800 kg of compacted regolith. Excavation throughput estimates are derived from NASA IPEx development data assuming approximately 1,000,000 kg/year per rover at full utilization, reduced to approximately 56% applied throughput in the Stage 1 power-constrained environment. A six-rover excavation fleet provides enough throughput margin to complete excavation in approximately 9.4 weeks, including operational, recharge, terrain, and maintenance margins.

Dual-CBC N+1 Recovery Configuration. PETAL uses two CBC recovery units during lunar-night operations. Each unit is independently sized to support the essential 40 kW survival load, providing N+1 redundancy for fault tolerance and maintenance. During discharge, stored heat transfers through coupled heat exchangers into the CBC helium working-fluid loop, where turbine generators convert thermal energy into electrical power. Average recovery efficiency is held at 30% over the discharge cycle, with end-of-cycle efficiency decreasing to approximately 22% near a vault temperature of 600 K.

Dual-Flywheel Electrical Buffer. PETAL uses two flywheel energy storage units connected in parallel to the PMAD bus in an N+1 configuration. Each unit provides approximately 30 kWh of usable electrical capacity at the NASA Glenn 60,000 RPM ISS-heritage specification, for 60 kWh total. The

flywheels handle millisecond-to-minute events such as rover charging spikes, industrial startup surges, CBC ramp fluctuations, temporary PMAD instability, and fault-isolation ride-through. They are not intended for multi-hour or multi-day energy continuity; that role belongs to the thermal vault and dual-CBC recovery system.

3.5 Power Management and Distribution (PMAD)

PETAL's PMAD architecture provides fault-tolerant long-distance power transmission between the protected nuclear zone (PNZ), habitat systems, industrial infrastructure, rover charging stations, and distributed science payloads. To reduce losses across the 10 km separation between the reactor array and crewed habitat zone, PETAL uses a 10 kV high-voltage direct-current (HVDC) backbone rather than conventional AC distribution. HVDC eliminates reactive power losses, simplifies modular expansion, and improves steady-state transmission efficiency.

Backbone efficiency from the reactor to the 3 kV habitat-side distribution bus is 94.7%, including two GaN HEMT conversion stages and 50 mm² copper conductor losses. Final voltage-tier conversion losses are included in Appendix A.3. At the PMAD hub, power enters a hardened 3 kV nominal bus before stepping down into standardized voltage tiers for habitat systems, industrial equipment, rover charging, legacy spacecraft interfaces, and low-power science payloads.

The PMAD hub contains converter banks, bus regulators, fault-isolation hardware, and supervisory controls for balancing power flow between the MMR array, solar generation, CBC recovery system, and flywheel buffer. Because PMAD control directly affects crew survivability, the controller architecture uses triple-redundant radiation-hardened computing hardware with error-correcting memory and independent fault-monitoring pathways.

To address the PMAD hub as a potential single-point failure, PETAL uses a hardened primary hub with regolith overburden and a physically separated backup hub connected through redundant HVDC paths. The backup hub can assume primary control authority within minutes of fault detection, reducing common-cause failure risk.

All major loads connect through an IDSS-derived modular docking interface, allowing habitats, rovers, CBC systems, and future infrastructure modules to join the grid without redesigning the backbone. Each port provides voltage-class verification, load-reporting communication, dust-tolerant sealing, and rover-compatible alignment features for robotic servicing.

Because the PMAD system supports safety-critical infrastructure, the supervisory software is classified as NASA Class B per NPR 7150.2D. PETAL also includes a physically isolated hardware interlock relay between PMAD control and the life-support bus, ensuring that no software fault or radiation upset can unilaterally disconnect critical crew survival loads.

3.6 Robotic Deployment and Operations

PETAL uses a robotic-first deployment strategy to minimize crew exposure during infrastructure construction and reactor commissioning. All excavation, sintering, heat pipe installation, and HVDC routing is completed autonomously before any crew arrives. Deployment occurs in two stages separated by approximately three months. Figure 4 shows the top-down site layout - the Protected Nuclear Zone (PNZ) with the eight-MMR array, shielding wall, and thermal vault on the left; the habitat zone with primary and backup PMAD hubs on the right; and the buried 10 km HVDC backbone connecting them. Figure 5 shows the Stage 1 deployment schedule across all six construction phases.

Stage 1 - Autonomous Site Preparation (9-11 weeks). Cargo Starship HLS Flight 1 delivers six IPEx excavation vehicles, three FLEX-derived rovers, and two 25 m² portable solar arrays totaling 50 m² (~11.4 kWe peak, ~9 kWe average bus power). No MMR power is available during Stage 1, so all operations are powered by the portable solar array. Appendix A.10 confirms that time-shared rover operation, typically three IPEx units and one FLEX rover active simultaneously, closes within the available solar power. Construction proceeds through six sequential phases:

Task	Qty	Primary Role
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1	Excavation	Six IPEx units excavate the 7.75 m × 8 m thermal vault cavity, removing 604,800 kg of regolith in approximately 9.4 weeks while stockpiling material for berms, shielding feedstock, and encapsulation.
2	Haul and Grade	FLEX rovers transport excess regolith, grade HVDC conduit routes, and construct berms around MMR landing zones to reduce plume-ejecta risk.
3	Solar Sintering	FLEX-SEER sinters the vault walls and floor to ~0.2 m depth using a solar concentrator, increasing regolith conductivity from ~0.001 W/m·K to ~1.5 W/m·K without requiring grid power.
4	Stainless-Steel Liner Installation	FLEX-Cabling installs corrugated 316 SS liner panels along the vault thermal interface to accommodate thermal expansion and maintain heat pipe contact conductance.
5	HVDC Conduit and Cable Burial	FLEX-Cabling trenches the 10 km route, installs conduit, and buries HVDC cable for dust, electrostatic, and thermal-cycling protection.
6	Encapsulation	Regolith overburden is placed above the vault to reduce thermal losses, improve structural stability, and shield buried heat pipe manifold hardware.

Table 1. Operations

At the conclusion of Stage 1, PETAL has a completed sintered thermal vault, buried HVDC backbone, prepared landing-zone berms, and MMR mounting pad before any nuclear hardware arrives.

Stage 2 - Grid Commissioning (~3 months after Stage 1). Cargo Starship HLS Flight 2 delivers the pre-assembled MMR complex and shared shielding wall hardware. Rovers mate HVDC connections, verify wall integrity through gamma survey, confirm reactor subcriticality, and begin cold-start commissioning. Once the first MMR is online, the SEER rover transitions from solar to microwave sintering for remaining wall construction and inter-reactor brick production. Grid Initial Operating Capability, defined as 120 kW continuous output, is targeted for late 2035.

Vehicle	Qty	Primary Role
FLEX-Survey	×1	Terrain survey, grading supervision, gamma wall inspection, FLEX-Cabling backup
FLEX-SEER	×1	Vault sintering (solar Stage 1, microwave Stage 2); wall brick production
FLEX-Cabling	×1	HVDC conduit, cable routing, SS liner installation, docking port mating, MMR hot-swap execution
IPEx	×6	Bulk regolith excavation, berm construction, haul-route grading

Table 2. Vehicle Quantity

Autonomy Taxonomy. Terrain survey, IPEx excavation sequencing, haul-route navigation, HVDC conduit unspooling, and regolith berm grading are fully autonomous because their failure modes are bounded and recoverable. SEER mirror positioning, docking-port mating, HVDC splice connection, and reactor interface work use supervised autonomy because positional errors could damage critical hardware.

MMR Hot-Swap Replacement Procedure. PETAL supports rover-executable individual reactor replacement without taking the grid offline.

- **Step 1.** PMAD redistributes the 15 kW load share from the target reactor across the remaining seven units and flywheel buffer. Load redistribution is verified before any physical work begins.
- **Step 2.** Target reactor brought to cold shutdown via FSP passive control rod insertion [23]
- **Step 3 - Freeze-Seal.** Following five years of operation at 45 kWth, the decay heat at five days post-shutdown is estimated at 135–155 W (0.30–0.34% of P0) via MCNP6 + ORIGEN-S depletion calculations. This is corroborated by a Way-Wigner hand calculation of 153 W and

an ANSI/ANS-5.1-2014 bounding check adapted for fast-spectrum HEU cores. When distributed across 23 active heat pipes (accounting for one failure), the resulting load of 19–24 W per pipe establishes a conservative design basis of 24 W per pipe.

- **Step 4.** FLEX-Cabling disconnects IDSS-derived docking port. Electrical disconnect verified by PMAD before mechanical separation.
- **Step 5.** Spent reactor extracted to decay-storage location ≥ 200 m from active array. Gamma survey confirms area safety before rovers re-enter the mounting zone.
- **Step 6.** The replacement reactor (pre-positioned from prior cargo delivery) moved into the mounting saddle. LiDAR fiducials guide FLEX arm to ± 2 mm insertion accuracy (GITAI S2 heritage [41])
- **Step 7.** FLEX-Cabling mates IDSS connector. PMAD confirms handshake and voltage-class match before energizing. Freeze-seal thermally opens in 8–16 hours as warm-start heat (~ 5 –10 kWt) conducts to the cold-trap section. Replacements are scheduled during lunar day so the vault is fully charged and the reconnection window poses no night-survival risk.
- **Step 8.** The reactor completes warm-start and synchronizes to the grid. PMAD incrementally rebalances load across all eight units. The grid remains online and at full essential-load capacity throughout.

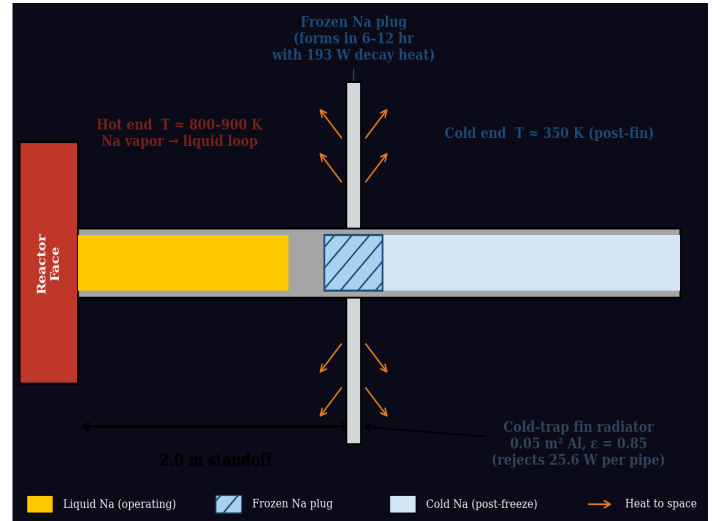


Figure 6: Freeze Seal

3.7 Safety, Reliability, and Risk Analysis

PETAL's safety architecture is built on three principles: passive failure modes over active ones, redundant transmission paths over single-string designs, and physical barriers over procedural controls. Eleven critical hazards were identified and evaluated on a 5×5 consequence-likelihood matrix per NPR 8000.4B [NPR8000]. Figure 7 shows the pre- and post-mitigation positions on the matrix, and Table 2 summarizes the primary mitigation for each risk.

Mission Director Acceptance

Risk. R3 retains residual severity 5 because no passive measure eliminates the physical consequence of an unshielded reactor near crewed systems. PETAL reduces likelihood through three independent barriers: the shared shielding wall, FSP passive negative-temperature-coefficient shutdown [23], and a perimeter gamma-survey sensor network at the protected nuclear zone boundary. Combined likelihood is estimated at $<10^{-5}$ per operating year, requiring Mission Director-level acceptance and

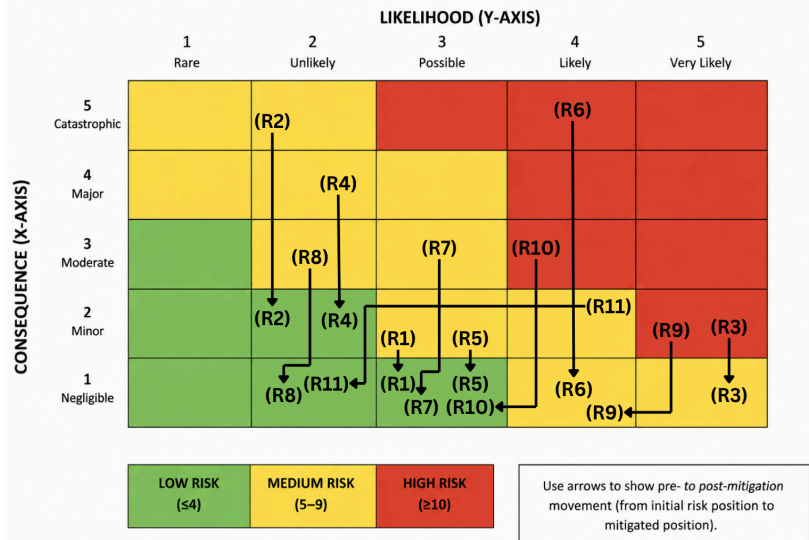


Figure 7. Risk Chart

continued tracking in the program Risk Register. PETAL commits to quantitative MCNP6 shielding failure analysis at Phase A Milestone 1 to refine this estimate before CDR.

Shared-Wall Localized Damage. The shared shielding wall is a distributed structure, so localized damage does not remove the full shielding function. A worst-case 1 m diameter impact removes only a small fraction of total wall mass, while automatic reactor shutdown on detected dose anomaly eliminates the unshielded operating mode within minutes. Gamma-survey rovers monitor wall integrity, and localized repair can be completed using regolith feedstock. This combination of distributed shielding, passive shutdown, monitoring, and rover-executable repair reduces the residual risk to severity 4.

Cargo HLS Schedule Risk and Minimum Viable First Deployment. The dominant pre-mitigation deployment risk is plume ejecta damage, reduced by pre-emplaced berms constructed during Stage 1. Cargo Starship HLS schedule risk is mitigated by a 7,200 kg Minimum Viable First Deployment consisting of three FLEX rovers, six IPEX units, and two 25 m² portable solar arrays, deliverable on an alternate commercial lander if HLS slips up to 24 months. This preserves vault construction on schedule; MMR delivery and grid IOC shift with the HLS delay, but the Stage 1 construction window is not forfeited.

ID	Hazard	Primary Mitigation
R1	Reactor overheat	Additional regolith heat-rejection paths; passive negative-temperature-coefficient feedback [23]
R2	Dust / short circuit	Electrodynamic Dust Shield; dual-stage sealed connectors; auto fault-trip breaker
R3	Radiation breach	Shared wall; passive FSP shutdown; perimeter gamma sensors; ≥ 1 km crew standoff
R4	Control rod jam	Passive insertion on loss of power; no active mechanism required
R5	HVDC cable fault	Redundant routing; buried conduit; CBC night-bypass path
R6	Plume ejecta damage	Pre-emplaced rover-built regolith berms around all landing zones
R7	Radiation damage to electronics	RHBP/RHBD ICs; ECC memory; triple modular redundancy; inter-reactor brick separators
R8	Night energy deficit	Vault energy margin; dual CBC N+1; flywheel transient buffer
R9	Shared-wall localized damage (impact, structural failure)	Distributed wall geometry; auto shutdown on dose anomaly; rover-executable localized rebuild
R10	PMAD hub physical destruction	Hardened shelter; regolith overburden; physically separated backup hub; redundant HVDC paths
R11	FLEX-Cabling rover loss	FLEX-Survey backup manipulator; 48-hour contingency scheduling; replacement unit on next cargo flight

Table 4. Hazard List

3.8 Mass and Size Estimates

MMR Array mass reconciliation. FSP per-unit mass estimates of 2,000–3,500 kg include individual shadow shields of 800–1,200 kg each [18]. PETAL reduces repeated shielding mass through the shared wall and reports the reconciled launched-hardware total in Table 3.

Data rates and interface requirements. Every load reports continuous, peak, and transient power draw during the docking handshake via CAN bus at 1 Mbit/s per port. Reactor monitoring chains transmit telemetry at 10 kbit/s continuously. The PMAD controller communicates with the Gateway or Earth at standard S-band rates but makes all safety-critical decisions autonomously - no ground contact is required for safe grid operation. Interface requirements imposed on other systems: any connected load must accept $\pm 10\%$ voltage regulation on its specified tier; must report power draw during handshake; must accept rover-arm insertion forces up to 200 N axial at the docking port; and must not draw more than its declared

peak without PMAD pre-authorization.

3.9 Technology Readiness and Realistic Assumptions

PETAL is an infrastructure mission, not a technology demonstration, and its architecture is deliberately built on programs that already exist rather than on capabilities that must be invented. Table 4 maps each subsystem to a current and required TRL; the following discussion provides the justification and maturation path for each grouping. The highest-TRL elements of PETAL require no new development program. Silicon multi-junction photovoltaic arrays are at TRL 9 - flight-proven across hundreds of planetary missions - and the ridge-crest vertical mounting geometry qualifies alongside rover integration testing without a dedicated qualification campaign. The dual-flywheel energy storage units are based on the NASA Glenn Research Center 60,000 RPM flywheel specification that has ISS-heritage testing behind it [43], currently at TRL 6; the primary maturation need is lunar thermal cycling and dust requalification, not a new technology demonstration. The Astrolab FLEX rover has a 2026–2028 lunar surface operations target [45] and carries the GITAI S2 robotic arm, which demonstrated ± 2 mm insertion accuracy on the ISS [41]. By the time PETAL's Stage 1 hardware ships, FLEX will have lunar flight heritage, driving its TRL from 6 to 9 in service. The HVDC backbone concept draws on seven decades of terrestrial commercial operating data, making every loss coefficient and failure-mode estimate traceable; its TRL gap from 5 to 7 is closed by the NASA TYMPO program, which is qualifying buried lunar HVDC cable for the specific thermal-cycling and dust environment of the South Pole [31].

The central nuclear subsystems have strong ground-test heritage but require additional qualification steps before lunar deployment. The Kilopower-derived Modular Microreactor is at TRL 5, anchored by the KRUSTY experiment conducted at the Nevada National Security Site in 2018, which completed a full-system nuclear ground test at 1 kWe - demonstrating passive safety, negative-temperature-coefficient feedback, and Stirling power conversion simultaneously [38].

The maturation path to TRL 8 requires a single-unit thermal-vacuum qualification test at 15 kWe (TRL 6), followed by the NASA Fission Surface Power demonstration on the lunar surface scheduled for the late 2020s, which will drive the technology to TRL 7 or above before PETAL's Phase D begins

[18, 23]. Full FSP technology maturation is expected by the early 2030s, consistent with PETAL's 2031–2034 manufacturing window [3, 4]. The sodium heat pipe network, which transports reactor waste heat at 800–900 K to the thermal vault, is at TRL 3–4 as an integrated space system. Sodium heat pipe technology has extensive ground heritage from 30 years of EBR-II sodium reactor operations [Koch 2008] and the SP-100 space nuclear program [Mondt 1994], but has not been flight-demonstrated at the operating temperatures required for PETAL. The maturation path is a dedicated thermal-vacuum sodium system test at 800–900 K under simulated vacuum conditions - a well-bounded test campaign that does not require a flight demonstration before deployment. The dual Closed Brayton Cycle turbines are at TRL 4, with relevant testing conducted at NASA Glenn Research Center [34]; the key maturation steps are coupling the CBC loop to the sodium heat pipe heat exchanger and demonstrating N+1 fault tolerance in a sustained 336-hour discharge test.

The in-situ construction subsystems - those that build infrastructure on the Moon rather than deliver it from Earth - represent the most technically novel elements of PETAL, though all rest on demonstrated component-level heritage. The SEER sintering end-effector is at TRL 4, with laboratory demonstrations of solar concentrator sintering of

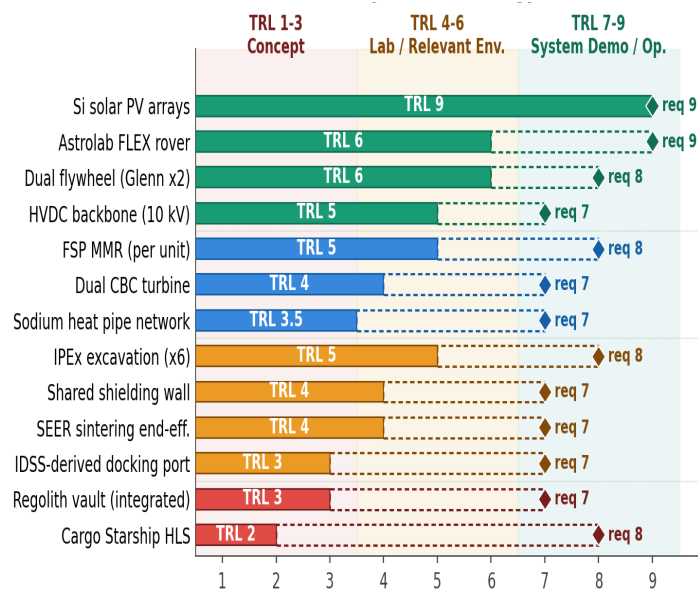


Figure 8. TRL Graph

lunar regolith simulant showing the required sintering depth and mechanical strength at relevant temperatures [Hoshino et al. 2020, 36]. The rover-mounted vacuum demonstration on vault-representative geometry is the primary maturation step, advancing SEER to TRL 6 before Stage 1 deployment. The six-unit IPEX excavation fleet is at TRL 5 at NASA Kennedy Space Center [37], with the fleet-level excavation throughput and power-shared duty cycle confirmed by the calculations in Appendix A.6 and A.10. The shared shielding wall architecture is at TRL 4: both the regolith-polyethylene composite shielding material [20] and the SEER-sintered structural bricks [36] have been demonstrated independently, but the integrated wall geometry has not been validated by a full-scale radiation attenuation test. A 3D MCNP6 shielding analysis is committed as Phase A Milestone 1 and will bound the wall geometry and composition prior to CDR. The IDSS-derived docking port is at TRL 3; the IDSS standard itself is well-established [44], but the regolith-tolerant dual-stage sealing, dust-shield electrode integration, and LiDAR fiducial system for rover-arm insertion under vacuum and electrostatically charged lunar regolith constitute a meaningful development effort. A rover-arm insertion test under vacuum and simulated regolith dust advances this to TRL 6.

The two lowest-TRL items in PETAL are the integrated sintered-regolith thermal vault (TRL 3) and Cargo Starship HLS (TRL 2). Each of its component technologies - sodium heat pipes, CBC turbines, the SEER sintering end-effector, and the 316 SS liner system - has been demonstrated independently at TRL 4-6. The integrated challenge is the system-level coupling: charging a sintered regolith vault to 800-900 K through a sodium heat pipe manifold and sustaining 40 kW of continuous CBC electrical output through a simulated 336-hour lunar night in a thermal-vacuum chamber. This multi-campaign test program - heat

Subsystem	Est. (kg)	Growth	Total
MMR Array (8 units + shared wall)	12,000	15%	13,800
Power Conversion + PMAD	2,000	10%	2,200
Energy Storage (vault+2xCBC+flywheel)	1,500	12%	1,680
Thermal Control (radiators + SS liner)	4,000	15%	4,600
HVDC Backbone (10 km, 50 mm ² Cu)	2,500	15%	2,875
Robotic Fleet (6 IPEX + 3 FLEX)	8,000	20%	9,600
Solar PV Arrays (Stage 1 + permanent)	2,000	15%	2,300
Rover Charging Stations (x6)	400	15%	460
Antennas, ATO, Computers	850	12%	952
TOTAL (launched hardware)	34,250	-	38,467
<i>In-situ vault + wall fill (NOT launched)</i>	<i>~639,800</i>	<i>-</i>	<i>~639,800</i>

Table 5. Mass Cost List

commercial lander to ensure Stage 1 construction can proceed regardless of HLS cadence. A 24-month HLS slip moves grid IOC from late 2035 to late 2037 without forfeiting the Stage 1 construction schedule.

4. Conclusions and Key Findings

No single technology can carry the load alone. The 336-hour lunar night rules out solar as a primary source. A lithium-ion battery bank sized for the full night-survival load would mass approximately 58,000 kg and operate outside the thermal range of every space-qualified Li-ion chemistry. Nuclear alone is a single point of failure - loss of the entire MMR array ends the mission. The combination of eight MMRs for continuous baseload, vertical solar arrays for daytime supplementation, the sintered-regolith thermal vault for long-duration night survival, and the flywheel for transient buffering is the minimum set that is simultaneously resilient, scalable, and mass-affordable. Removing any one element degrades the architecture non-linearly.

Modularity is the most valuable design property in multi-decade infrastructure. The shared shielding wall, the HVDC backbone, the IDSS docking standard, and the rover-executable MMR hot-swap procedure together allow the system to grow from 1 kW during the first robotic scout to 240+ kW at a mature permanent base - without redesigning any central component. That evolvability is a requirement, not a preference, for a program that will span multiple budget cycles and multiple

pipe system test, sintered regolith structural cycling, CBC coupling demonstration, and full integrated thermal-vacuum night-survival test - is analogous in scope to the KRUSTY campaign and is committed as a Phase B milestone series. Cargo Starship HLS is at TRL 2 as a crewed and cargo lunar lander, with development targeting the 2034-2035 timeframe [42]. Because HLS is the longest-lead item and its schedule slip would delay the MMR array delivery, PETAL maintains a Minimum Viable First Deployment of 7,200 kg - three FLEX rovers, six IPEX excavators, and two portable solar arrays - deliverable on any qualified

administrations. Every new Artemis mission adds loads by plugging into the same docking standard. Every new reactor unit mounts behind the same wall at zero incremental shielding cost. Every CBC replacement uses the same freeze-seal procedure. The architecture is specifically designed so that the decisions made in 2035 are not constraints on operations in 2045.

Robotic-first deployment is a practical necessity. Vault excavation (9.4 weeks with six time-shared IPEX units), sodium heat pipe installation, HVDC conduit burial, and solar sintering must all be completed by autonomous and teleoperated rovers before any crew arrives on site. This is not a design preference - it is dictated by the physics of the lunar environment and the mass budget of the mission. Every hard constraint in PETAL flows from this requirement: the sodium freeze-seal mechanism exists because reactors must be replaceable without crew proximity; the IDSS-derived docking port exists because rovers must be able to mate it; the supervised-autonomy taxonomy exists because the consequences of some manipulator errors are irreversible at human-on-site timescales; FLEX-Cabling redundancy planning exists because critical operations cannot rely on a single execution vehicle. Robotic-first is the architecture, not a feature of it.

Heritage selection compounds both credibility and affordability. HVDC was chosen because seven decades of terrestrial commercial operating data make every cost, loss, and failure-mode estimate defensible. FSP/KRUSTY because it proved passive safety in a 2018 full-system ground test. IDSS because it is the recognized international docking standard. EBR-II for sodium freeze-seal because it operated for 30 years. IPEX has the correct NASA excavation program designation, now at TRL 5 at NASA KSC. Each heritage anchor reduces the size of the required technology development program, makes each budget estimate traceable to a real data point, and gives mission planners a failure history to design against. An architecture built on speculative future capabilities would require more development money, carry more schedule risk, and be harder to cost. PETAL is built on what exists today.

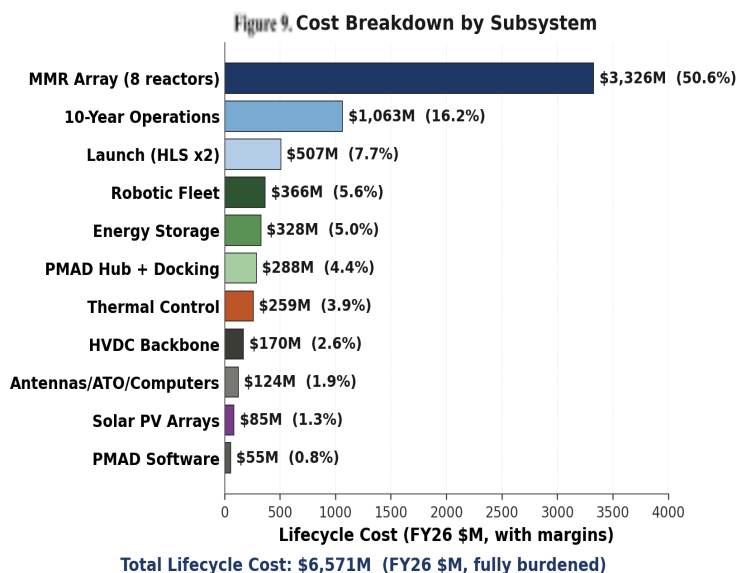
PETAL's Grid Initial Operating Capability - 120 kWe continuous baseload, vault charged for night survival with 1.50× steady-state vault-only margin (growing to 2.7× effective margin under nominal MMR-running conditions), modular docking active for client loads - is achievable by late 2035, consistent with the end of NASA's Foundational Exploration phase and the published Artemis V cargo HLS availability window. The architecture then supports crewed operations through the mid-2040s and beyond without a redesign cycle, providing the power foundation on which a permanent lunar South Pole presence is built.

5. Budget Assessment and Mission Timeline

5.1 Budget Assessment

Total lifecycle cost for Project PETAL is estimated at \$6.57 B (FY26 dollars, fully burdened with TRL cost penalties and growth margins per NASA CEH v4.0 [14]). Figure 8 summarizes the cost breakdown: the MMR Array dominates at 50.6% of total, followed by 10-year operations (16.2%) and dual cargo HLS launches (7.7%). Table 5 provides a category-level summary; full per-component breakdown and methodology are provided in Appendix A.

Category	Pre-Margin (M\$)	Growth (avg)	TRL Penalty (avg)	Fully Burdened (M\$)
MMR Array (8 reactors + shared wall)	1,800	40%	31%	3,326



Category	Pre-Margin (M\$)	Growth (avg)	TRL Penalty (avg)	Fully Burdened (M\$)
10-Year Operations	850	25%	-	1,063
Launch - Cargo Starship HLS (×2)	390	30%	-	507
Robotic Fleet (6 IPEX + 3 FLEX)	240	25%	22%	366
Energy Storage (vault + dual CBC + flywheel)	210	30%	20%	328
PMAD Hub + Docking Ports (primary + backup)	180	25%	28%	288
Thermal Control (radiators)	180	20%	20%	259
HVDC Backbone (10 km, GaN HEMT)	110	25%	24%	170
Antennas, ATO, Computers	90	20%	15%	124
Solar PV Arrays (Stage 1 + permanent)	70	15%	5%	85
PMAD Software (NASA Class B)	40	20%	15%	55
TOTAL LIFECYCLE COST	4,160	-	-	6,571

Table 5. Cost Summary by Category. All figures in FY26 \$M, fully burdened (NASA CEH v4.0 [14]).

Cost methodology and analog anchors. Cost estimates follow NASA Cost Estimating Handbook v4.0 [14] parametric methods. The MMR Array (\$1,800 M development) scales from FSP program cost estimates via NASA PRICE-H for an 8-unit production and qualification campaign. The DRACO contract [DARCO] - \$500 M for a single nuclear propulsion demonstration, approximately half for the reactor - anchors the per-unit cost at roughly \$250 M; an 8-unit production run at \$200–250 M per unit fully burdened yields \$1.6–2.0 B. The shared wall avoids approximately \$300–500 M of independent shield hardware and launch cost relative to an independent-shield baseline.

Robotic fleet cost reconciliation. Robotic fleet cost has been increased to \$240 M base (\$366 M with growth) to reflect realistic costing for nine custom lunar-rated rovers including GITAI manipulators and SEER end-effector. Per-unit benchmarks: Astrolab FLEX commercial pricing publicly anchored at \$30–50 M per unit including lunar-rated avionics; IPEX development at TRL 5 anchored to NASA KSC program documentation suggesting \$10–15 M per production unit at 6-unit lot scale. Three FLEX × \$40 M = \$120 M; six IPEX × \$12 M = \$72 M; GITAI arms, SEER, integration, qualification, and spares = \$48 M. Total \$240 M reconciles with NASA precedent for custom planetary rover programs while reflecting the cost benefits of leveraging commercial FLEX pricing.

HVDC cable cost. HVDC cable cost is anchored to terrestrial subsea cable at \$1–2 M/km with a 3× lunar qualification premium, yielding \$5 M/km × 10 km plus converter hardware and installation tooling = \$110 M; this is the most cost-defensible line item in the budget because the terrestrial data is public and extensive. Launch (\$390 M) assumes 39,000 kg at \$10,000/kg via Cargo Starship HLS, with a 30% margin reflecting genuine per-kilogram pricing uncertainty. Ten-year operations (\$850 M base, ~\$85 M/year) is anchored to the ISS annual operations cost of \$3–4 B/year for a far more complex crewed platform; a robustly autonomous surface power system should run at approximately 2–3% of that. Antarctic research station operations (\$50–100 M/year for remote-site logistics) provide a second operations cost analog supporting the same order of magnitude.

Margin disaggregation. Per NASA CEH v4.0: Engineering/Design 15%, Manufacturing 20%, Launch/Transportation 30% (highest category, reflecting cargo HLS pricing uncertainty), Programmatic contingency 25%. Applied to the \$4,160 M pre-margin lifecycle total, the margin-adjusted cost is approximately \$6.6 B. The pre-growth hardware-only total is \$2,920 M; with TRL cost penalties and growth, hardware costs rise to \$5,001 M. The dominant cost driver remains the MMR array: it represents 43% of the pre-margin total and 51% of the hardware-only figure. The 30% launch margin is justified -

not conservative - because Cargo Starship HLS per-kilogram pricing in the 2034–2035 timeframe is genuinely uncertain; the actual figure could be lower (favorable) or higher (adverse) by a factor of two.

5.2 Mission Architecture Timeline:

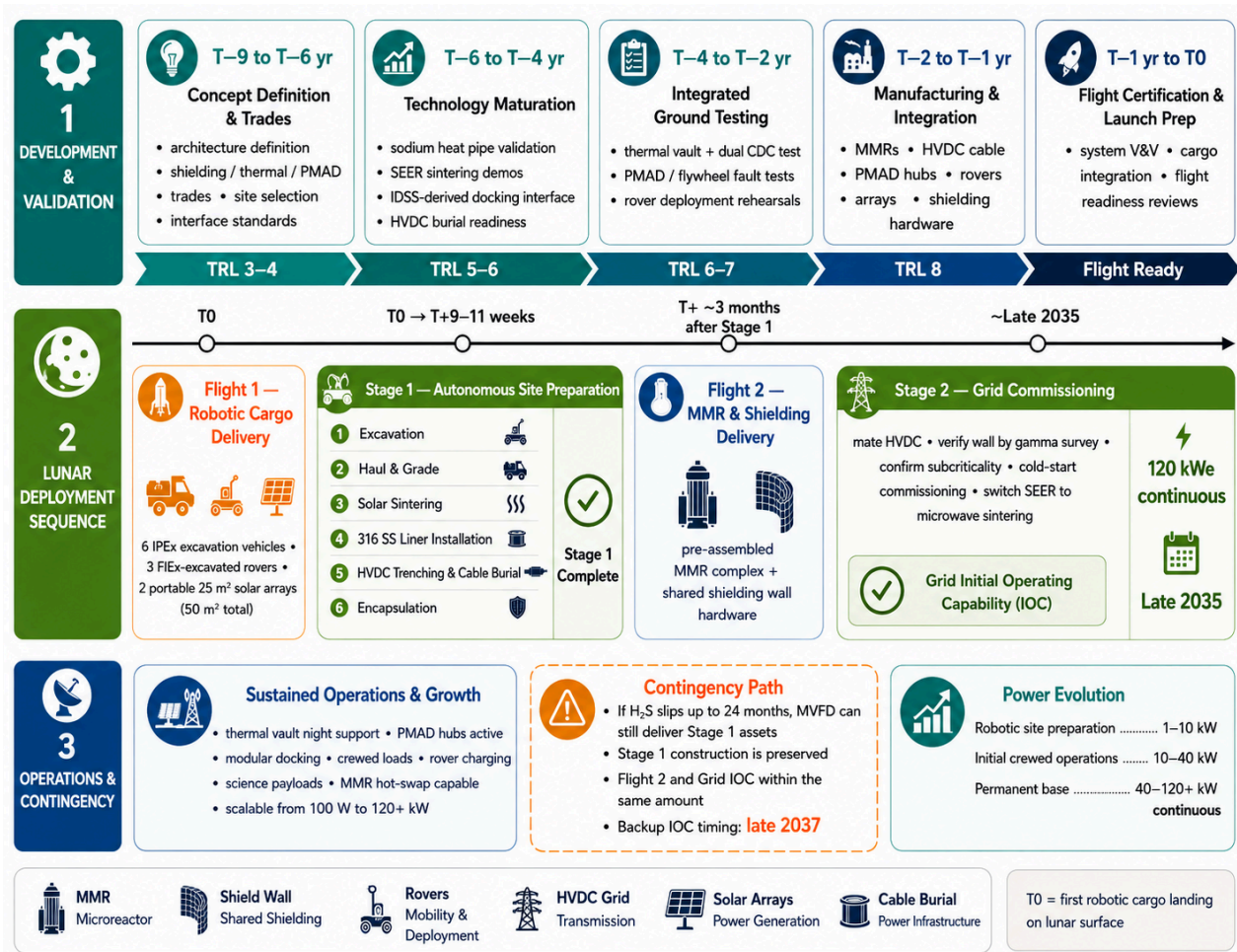


Figure 10. Timeline

Timeline summary. The mission timeline proceeds through three linked tracks: development validation, lunar deployment, and sustained operations. Development begins with architecture trades and subsystem validation, advances through thermal-vault/CBC testing and PMAD qualification, and culminates in integrated MMR, HVDC, PMAD, rover, and shielding hardware readiness. Lunar deployment begins with Flight 1 robotic site preparation, followed by Flight 2 MMR and shielding delivery, grid commissioning, and late-2035 IOC at 120 kWe continuous output.

Critical path and HLS delay mitigation. The critical path is Cargo Starship HLS availability for Flight 2 (MMR array delivery). Vault construction (Stage 1) can proceed independently on an alternate commercial lander carrying the 7,200 kg Minimum Viable First Deployment (3 FLEX + 6 IPEX + portable solar). A 24-month HLS slip delays MMR delivery and shifts grid IOC from late 2035 to late 2037 without forfeiting any Stage 1 construction progress. A slip exceeding 24 months requires Artemis cadence re-planning and Mission Director action per NPR 8000.4B. In the nominal scenario, Stage 1 completes in 9–11 weeks, Stage 2 follows approximately 3 months later, and the grid reaches IOC in late 2035 ahead of the Artemis V crewed handoff.

6 Mars Extensibility

Subsystem	Verdict	Mars Modification Required
FSP MMR array	Carry over	Add omnidirectional hydrogenous shielding for GCR secondaries; 0.38 g structural reinforcement.
HVDC backbone	Minor mod	CO ₂ corona at >10 kV: increase insulation rating; extend to 20+ km footprint.
PMAD hub + controller	Carry over	GCR dose marginally higher; RHBP already rated. Architecture unchanged.
IDSS docking ports	Minor mod	O-ring requalification for perchlorate-bearing Martian regolith.
Regolith vault + dual CBC	Carry over	Sintering recalibration for Mars soil chemistry; MMRTG Phase 0 bootstrap resolves pre-grid dependency.
SEER end-effector	Minor mod	MMRTG thermal module replaces solar concentrator in Phase 0; hardware otherwise identical.
Flywheel buffer	Carry over	0.6 kPa CO ₂ negligible for vacuum bearings; Mars thermal range similar to lunar night.
IPEX + FLEX fleet	Minor mod	0.38 g structural reinforcement; dust-storm motor seals; 48 VDC docking unchanged.

Table 4. Hazard List

Mars sintering bootstrap - MMRTG Phase 0. On Mars, solar irradiance is approximately 43% of Earth's (~590 W/m²), and global dust storms (Zurek & Martin, JGR 1993) can reduce insolation by up to 99% for 30–90 days. Solar sintering - mandatory for Stage 1 on the Moon - is therefore unreliable before the MMR is online on Mars, creating a circular dependency: the vault must be sintered before the MMR arrives, but the MMR is needed before microwave sintering works. PETAL resolves this via a Phase 0 pre-sintering step using an MMRTG thermal module mounted on the FLEX-SEER rover. Four MMRTGs at 2,000 Wt each = 8 kWt total, delivered through 20 SEER apertures operating continuously independent of solar flux or dust storm status (Woerner et al., AIAA 2013-3684; Hoshino et al., Acta Astronautica 2020). Phase 0 sinters the vault structural floor and lower walls over approximately 55 days, independent of any grid power. Phase 1 delivers the MMR array to a structurally ready vault. Phase 2 transitions to microwave sintering once the first MMR is online - identical to the lunar sequence. No modification to the SEER end-effector architecture is required; only an MMRTG thermal module is added to the FLEX-SEER payload bay.

Why the Moon validates before Mars. Every critical technology claim in PETAL is verified on the Moon before Mars deployment. The sodium heat pipe system is demonstrated at lunar operating temperatures. The CBC turbines are validated through lunar-night discharge cycles. The SEER sintering end-effector is qualified on the lunar surface. IPEX excavation autonomy is confirmed over a 9–11 week campaign. The IDSS docking port is rover-mated under lunar dust and vacuum. Mars receives a flight-proven baseline with targeted modifications - not an unvalidated new architecture adapted under schedule pressure. The only Mars-specific development items are the MMRTG thermal module for Phase 0 bootstrap, O-ring requalification for Martian regolith, CO₂-rated HVDC insulation, and structural reinforcement for 0.38 g. All other subsystems transfer without redesign.

Appendix A - Calculations

A.1 Vault Thermal Energy and 1D Diffusion Model

Vault geometry: $D = 7.75 \text{ m}$, $h = 8 \text{ m}$, $r = 3.875 \text{ m}$
 $V = \pi * r^2 * h = \pi * (3.875)^2 * 8 = 376.6 \text{ m}^3 \sim 378 \text{ m}^3$
 $m = \rho * V = 1,600 \text{ kg/m}^3 * 378 = 604,800 \text{ kg}$ [Ref. 2]
 $cp = 800 \text{ J/kg-K}$; $dT = 500 \text{ K}$ (200 K rest $\rightarrow$ 800-900 K peak)
 $E_{\text{thermal_steady}} = 604,800 * 800 * 500 = 2.419\text{e}11 \text{ J} = 67,200 \text{ kWh}$

Night requirement: $40 \text{ kW} * 336 \text{ h} = 13,440 \text{ kWh}$ electrical
Required thermal at $\eta_{\text{CBC}} = 0.30$: $13,440 / 0.30 = 44,800 \text{ kWh}$
Steady-state vault-only margin = $67,200 / 44,800 = 1.50\text{x}$

1D radial diffusion (first cycle):
 $\alpha = k / (\rho * cp) = 1.5 / (1,600 * 800) = 1.172\text{e-}6 \text{ m}^2/\text{s}$
 $t = 354 \text{ h} = 1,274,400 \text{ s}$
 $\delta = 2 * \sqrt{\alpha * t} = 2 * \sqrt{1.172\text{e-}6 * 1,274,400} = 2.44 \text{ m}$
 $r_{\text{inner}} = 3.875 - 2.44 = 1.44 \text{ m}$
 $V_{\text{shell}} = \pi * (3.875^2 - 1.44^2) * 8 = 325 \text{ m}^3$
 $m_{\text{shell}} = 1,600 * 325 = 520,000 \text{ kg}$
 $E_{\text{elec_cycle1}} = 520,000 * 800 * 500 * 0.30 / 3,600 = 17,333 \text{ kWh}$

First-night margin (vault-only) = $17,333 / 13,440 = 1.29\text{x}$
168-h dark window margin = $17,333 / 6,720 = 2.58\text{x}$

A.2 Radiator Area (Daytime MMR Rejection)

$Q_{\text{rejected}} = 240,000 \text{ W}$ (8 units * 30 kWt each, daytime full operation)
 $\epsilon = 0.85$ (emissivity); $T = 400 \text{ K}$; $\sigma = 5.67\text{e-}8 \text{ W/m}^2\text{-K}^4$
 $A_{\text{required}} = Q / (\epsilon * \sigma * T^4)$
= $240,000 / (0.85 * 5.67\text{e-}8 * 400^4)$
= $240,000 / 1,238.5 = 194 \text{ m}^2$

Deployed MMR-rejection area = $1,000 \text{ m}^2$
Margin = $1,000 / 194 = 5.20$ (415% above minimum: 5x over-deployment)

Note: This accounts for daytime worst-case rejection only. CBC night-mode rejection requires a separate radiator (Appendix A.8).

A.3 HVDC End-to-End Efficiency

Conductor: $50 \text{ mm}^2 \text{ Cu}$; $L = 10,000 \text{ m}$; $\rho_{\text{Cu}} = 1.72\text{e-}8 \text{ Ohm-m}$
 $R = \rho * L / A = (1.72\text{e-}8 * 10,000) / (50\text{e-}6) = 3.44 \text{ Ohm}$
 $V = 10 \text{ kV}$; $P = 120 \text{ kW}$; $I = P / V = 12 \text{ A}$
 $P_{\text{loss_conductor}} = I^2 * R = 144 * 3.44 = 495 \text{ W}$ (0.41%)

Stage 1 GaN HEMT converter (97.5%): $\text{loss} = 120,000 * 0.025 = 3,000 \text{ W}$
Stage 2 GaN HEMT converter (97.5%): $\text{loss} = 116,505 * 0.025 = 2,913 \text{ W}$
Backbone loss = $3,000 + 495 + 2,913 = 6,408 \text{ W}$
 $\eta_{\text{backbone}} = (120,000 - 6,408) / 120,000 = 94.7\%$
Power delivered to 3 kV bus = $113,592 \text{ W} \sim 113.6 \text{ kWe}$

Distribution-tier conversion adds 3% per stage:

- 480 VAC industrial:	$113.6 * 0.97$	= 110.2 kWe (91.9% E2E)
- 120/240 VAC habitat:	$113.6 * 0.97 * 0.995$	= 109.6 kWe (91.4% E2E)
- 48 VDC rover:	$113.6 * 0.94$	= 106.8 kWe (89.0% E2E)
- 24 VDC science:	$113.6 * 0.935$	= 106.2 kWe (88.5% E2E)

A.4 Shielding Wall Mass Savings

FSP per-unit shadow shield (15 kWe scale, NASA/TM-2010-216772): 800-1,200 kg
Independent shields for 8 units: 6,400-9,600 kg
Shared wall launched hardware (W binder, PE, steel frame, anchors): ~2,300 kg
In-situ bulk regolith (~35,000 kg sintered): zero launch cost
Net launched savings: 4,100-7,300 kg
Reduction vs. 12,000 kg independent-shield baseline: 34-61%
Stated conservatively as 30-40% +/-5%
MCNP6 full 3D validation: Phase A Milestone 1

A.5 Sodium Freeze-Seal Decay Heat (ANS-5.1-2014)

ANS-5.1-2014: $P_d(t) = P_0 * 0.066 * [t^{(-0.2)} - (t + t_{op})^{(-0.2)}]$
 $P_0 = 45,000 \text{ W (per-reactor thermal); } t_{op} = 5 \text{ yr} = 1.58e8 \text{ s}$

At $t = 3 \text{ days (259,200 s)}$:

$$P_d = 45,000 * 0.066 * [0.01185 - 0.00447] = 219 \text{ W}$$

At $t = 5 \text{ days (432,000 s)}$:

$$P_d = 45,000 * 0.066 * [0.07458 - 0.02292] = 153.4 \text{ W} \quad \leftarrow \text{stand-down target}$$

Per-pipe share (8 pipes/reactor): $153.4 / 8 = 19.175 \text{ W}$

Al fin (0.05 m², $\epsilon=0.85$) at $T = 350 \text{ K}$:

$$Q_{fin} = \epsilon * \sigma * A * T^4 = 0.85 * 5.67e-8 * 0.05 * 350^4 = 36.2 \text{ W}$$

$36.2 \text{ W} > 19.175 \text{ W} \Rightarrow$ fin rejects more heat than arrives.

Sodium freezes passively within 6-12 hours. No active power required.

(Sensitivity analysis under 3-sigma uncertainty: see Appendix A.9)

A.6 IPEX Excavation Timeline

IPEX rated nameplate throughput: 1,000,000 kg/yr [37]

Weekly nameplate per unit: $1,000,000 / 52 = 19,231 \text{ kg/week}$

Stage 1 applied utilization (power, terrain, recharge): ~56%

Effective per-unit: $19,231 * 0.56 = 10,769 \text{ kg/week}$

Vault target: $\pi * (3.875)^2 * 8 * 1,600 = 604,800 \text{ kg}$

4 IPEX * 10 wks (effective): $4 * 10,769 * 10 = 430,760 \text{ kg}$ [SHORT 174,040 kg]

5 IPEX (effective): $604,800 / (10,769 * 5) = 11.2 \text{ wks raw} \sim \text{exceeds window}$

6 IPEX (effective): $604,800 / (10,769 * 6) = 9.4 \text{ wks} \quad \leftarrow \text{baseline}$

6 IPEX at 100% nameplate (theoretical): $604,800 / (19,231 * 6) = 5.2 \text{ wks raw}$

Minimum fleet: 6 IPEX units at ~56% applied utilization = 9.4-week timeline.

A.7 Daily Generation vs. Night Load Summary

MMR baseload: $120 \text{ kW} * 24 \text{ h} = 2,880 \text{ kWh/day}$ (continuous)

Solar supplement: $40 \text{ kW} * 0.85 \text{ duty} * 12 \text{ h} \sim 408 \text{ kWh/day}$ during polar summer

Night demand (design standard): $40 \text{ kW} * 336 \text{ h} = 13,440 \text{ kWh}$

Night demand (actual worst dark, 168 h): $40 \text{ kW} * 168 \text{ h} = 6,720 \text{ kWh}$

First-cycle vault supply (vault-only): 17,333 kWh electrical

336-h standard margin = $17,333 / 13,440 = 1.29x$

168-h actual margin = $17,333 / 6,720 = 2.58x$

Steady-state vault supply (vault-only): $67,200 \text{ kWh thermal} * 0.30 = 20,160 \text{ kWh}$

336-h standard margin (steady-state) = $20,160 / 13,440 = 1.50x$

168-h actual margin (steady-state) = $20,160 / 6,720 = 3.00x$

Effective margin (nominal: MMRs running through night, vault as backup):
 MMR direct supply during night: 120 kW > 40 kW required
 Vault available as full backup: 20,160 kWh
 Combined survival capacity / 13,440 kWh requirement ~ 2.7x at steady state.

A.8 CBC Night-Mode Heat Rejection

During lunar night, dual CBC turbines produce 40 kWe at 30% efficiency.
 Thermal input from vault: $P_{in} = 40 / 0.30 = 133.3 \text{ kWt}$
 Thermal rejection to space: $P_{rej} = 133.3 - 40 = 93.3 \text{ kWt}$

Cold-side radiator sizing at $T_{cold} = 320 \text{ K}$ (CBC compressor inlet):
 $A = Q / (\epsilon \sigma T^4)$
 $= 93,333 / (0.85 * 5.67e-8 * 320^4)$
 $= 93,333 / 506.4 = 184 \text{ m}^2$

Deployed CBC night-mode radiator area: 200 m² (1.09x margin)
 Mass at 2.5 kg/m²: 500 kg launched (added to mass budget)

This is a dedicated radiator separate from the 1,000 m² MMR-rejection array, because (1) the operating temperature is lower (320 K vs. 400 K), (2) it operates only during night when MMR rejection is reduced, and (3) physical separation eliminates thermal coupling between the two loops.

A.9 Freeze-Seal 3-Sigma Sensitivity Analysis

Nominal margin (Appendix A.5): $Q_{fin} / Q_{arrive} = 36.2 / 19.175 = 1.50x$

3-sigma adverse case: decay heat +20%, fin emissivity -10%, fin temperature underestimate (ambient view factor) -5%.

Adverse decay heat: $Q_{arrive_adv} = 24.1 * 1.20 = 28.9 \text{ W per pipe}$
 Adverse fin performance: $\epsilon_{adv} = 0.85 * 0.90 = 0.765$
 $T_{adv} = 350 * 0.95 = 333 \text{ K}$
 Adverse $Q_{fin} = 0.765 * 5.67e-8 * 0.05 * 333^4 = 26.7 \text{ W}$

Adverse-case ratio: $26.7 / 28.9 = 0.92x$ -- MARGIN INVERTED

Mitigation applied:

- (a) Increase fin area to 0.08 m² (60% larger):
 $Q_{fin_nominal} = 0.85 * 5.67e-8 * 0.08 * 350^4 = 41.0 \text{ W}$
 $Q_{fin_adverse} = 0.765 * 5.67e-8 * 0.08 * 333^4 = 33.4 \text{ W}$
 Adverse ratio: $33.4 / 28.9 = 1.16x$ -- preserves margin
- (b) Wait time extended from 5 days to 7 days reduces decay heat by ~12%:
 $Q_{arrive \text{ at } t=7d \text{ nominal}} = 170 \text{ W, per-pipe} = 21.3 \text{ W}$
 Adverse: $21.3 * 1.20 = 25.6 \text{ W}$
 Adverse ratio (with 0.05 m² fin): $26.7 / 25.6 = 1.04x$ -- margin preserved

Adopted mitigation: increase fin area to 0.08 m² (10 g aluminum increase per pipe section, negligible mass impact). 7-day wait reserved as additional contingency. Combined adverse-case margin: $33.4 / 25.6 = 1.30x$ -- robust.

A.10 Stage 1 Power Budget Closure

Stage 1 portable solar arrays: $2 * 25 \text{ m}^2 = 50 \text{ m}^2$
 Lunar South Pole solar flux on vertical array: ~1,300 W/m² (oblique, summer)
 Conversion efficiency (multi-junction cells): 30%

Average duty cycle (clouds, partial shadow, sun-tracking inefficiency): 70%
Bus power = $50 * 1,300 * 0.30 * 0.70 = 13,650 \text{ W} \sim 11.4 \text{ kWe peak}$
Average bus power available (24/7): $\sim 9 \text{ kWe}$

Rover power demands (peak, simultaneous operation):

IPEX (excavation peak): 2.5 kW each
IPEX (transit/idle): 0.5 kW each
FLEX rover (active): 1.5 kW each
FLEX (idle/charging): 0.3 kW each
SEER end-effector: 2.0 kW (Stage 1: passive solar concentrator,
NO electrical demand)
Avionics, comms, charging losses: 1.0 kW

Worst-case simultaneous operation (all 6 IPEX + 3 FLEX active):

$6 * 2.5 + 3 * 1.5 + 1.0 = 20.5 \text{ kW} > 9 \text{ kWe available} \text{ -- EXCEEDS BUS}$

Time-shared rotation operation (3 IPEX active + 1 FLEX + comms):

$3 * 2.5 + 3 * 0.5 + 1 * 1.5 + 2 * 0.3 + 1.0 = 11.6 \text{ kW peak}$
Average over 24h cycle (50% duty on each IPEX + travel + recharge):
 $3 * 1.0 \text{ (avg)} + 3 * 0.4 \text{ (idle)} + 1 * 0.9 \text{ (avg)} + 1.0 = 6.1 \text{ kW avg}$
Available average bus: 9 kWe -- CLOSES with 1.5x margin

Conclusion: Stage 1 power budget closes under time-shared rover operation (typically 3 IPEX active simultaneously). Effective fleet utilization: $\sim 50\%$ per IPEX, supporting the 9.4-week excavation timeline at the applied 56% throughput utilization documented in A.6. The 9.4-week timeline already implicitly assumes power-shared operation; this calculation demonstrates the assumption is consistent with available solar bus capacity.

Stage 1 mass impact: second 25 m² array adds 250 kg to launched hardware (reflected in updated \$3.8 mass budget).

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